

Estimating inter-individual variability of dynamic energy budget model parameters for the copepod *Nitocra spinipes* from existing life-history data

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ABSTRACT

The degree of biological variability within a population is an important factor for its ecological success. Yet, individual-based population models (IBMs) that utilize the Dynamic Energy Budget (DEB) theory as a mechanistic basis to simulate an individual's life history, largely rely on rule-of-thumb estimates of inter-individual variability of their parameter values. In this study, we explored how data from previous life-history experiments with the copepod *Nitocra spinipes* could be used to make realistic estimates of variability in DEB parameter values for this species. We specifically investigated to which one of 12 possible parameters variability should be added, and what shape and scale its probability distribution should have, to most accurately reproduce the experimental data's variation with model simulations. The best result was achieved by drawing the parameter 'surface-area-specific maximum assimilation rate' from a log-normal distribution with a standard deviation of 0.15 or by drawing the maturity thresholds for birth and puberty (assuming proportionality between the two) from a log-normal distribution with a standard deviation of 0.40. In both cases, the model not only accurately reproduced the variation observed in development time data (which had been used in the fitting process) but also gave adequate predictions of the variation observed in brood size data (which had not been used for fitting). Although the findings of this study apply to one species only, the methods presented here are a proof of principle for how the question of intra-specific variability in a DEB model, for any species, can be addressed in a simple, yet purposeful manner.

1. Introduction

No two organisms are alike. Even among individuals of the same species genetic and phenotypic differences exist. While researchers often aim to link variations in observable traits to the fitness and the performance of individual organisms, a growing body of research has focused on how inter-individual variation affects the ecological success of a population as a whole (Agashe, 2009; Gamfeldt and Källström, 2007; González-Suárez and Revilla, 2013). In a comprehensive review on this subject, Forsman and Wennersten (2016) evaluated 45 experimental and comparative studies, 42 of which found a positive relationship between inter-individual variability and the ecological success of a population or species. Indicators of ecological success included: lower vulnerability to environmental changes, reduced population fluctuations, higher colonization success, larger geographic range distributions, and lower risk of extinction. Furthermore, multiple studies indicate that the population benefits of genetic and phenotypic variability were more prominent under more stressful conditions

(Caesar et al., 2010; Ganz and Ebert, 2010; Randall Hughes and Stachowicz, 2011).

Despite the apparent importance of inter-individual variability for the performance of a population, it is often disregarded in mathematical population models. Such models are used to simulate the population dynamics of a species under conditions that are impracticable to set up experimentally. In environmental decision-making, this can be helpful to assess untested and potentially harmful scenarios that may occur in the future, e.g., due to chemical contamination or climatic changes. A model type that is particularly suitable to incorporate inter-individual variability is the individual-based model (IBM) (Railsback and Grimm, 2019). Compared to other population models, IBMs have the advantage to keep track of individual organisms, each with its own set of parameters and state variables. Nevertheless, realistically reproducing variation found in life-history data of a species, such as the size and age at maturation or the number of offspring per brood, in an IBM, is not always straightforward. Especially if one life-history process is controlled by more than one model parameter or if, conversely, one

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model parameter influences several life-history processes, estimating the variability of each individual parameter from life-history data can be intricate.

In IBMs that employ the Dynamic Energy Budget (DEB) theory, the life-history trajectory of an organism depends on at least 12 physiological parameters. Within a set of fixed model equations that describe general rules of metabolic organization common to all organisms, these parameters account for all species-specific manifestations of life-history processes, such as food uptake, growth, development, and reproduction (Kooijman, 2010b). Given a set of experimental life-history data for a species, the software package DEBtool (Marques et al., 2018), which implements the “covariation method” presented by Lika et al. (2011), can be used to estimate the optimal parameters to reproduce this species’ life history in a DEB model. However, this optimal set of parameters only represents an ‘average’ animal of the species and does not take into account inter-individual variability. In recent years, many researchers from different disciplines have undergone training in DEB theory and parameter estimation (e.g., during the biannual DEB telecourses, symposia, and thematic schools), which has led to many new entries in the Add-my-Pet (AmP) data collection (2033 entries as of April 15, 2020) (AmP, 2020). While for some entries, data of multiple individual organisms were used to estimate a species’ DEB parameters, it is important to notice that all inter-individual variation was simply treated as data uncertainty even though it is likely that at least part of this variation is attributed to actual biological variability within the species. For applications of DEB theory in which the average performance of an organism is the main focus, this is no point of concern. For DEB-based IBMs, however, knowledge of how the physiological parameters vary among individuals of the same species is essential to realistically portray the variability within a population in model simulations.

Based on suggestions by Kooijman et al. (1989), Martin et al. (2012) introduced variability in DEB-IBMs by applying scatter to a single primary parameter, namely the surface-area-specific maximum assimilation rate $\{\dot{p}_{Am}\}$, and consequently all secondary parameters that depend on $\{\dot{p}_{Am}\}$. The scatter was added via a random number, log-normally distributed around a mean value of 1, drawn for each animal individually and multiplied with the default $\{\dot{p}_{Am}\}$. This method requires only one new model parameter, i.e., the standard deviation of the log-normal distribution, and introduces inter-individual variability in numerous life-history traits simultaneously. While we embrace this “one-parameter approach” for its simplicity and practicability, a sound rationale for why $\{\dot{p}_{Am}\}$, and no other parameter instead, should be subject to variation is currently lacking. Furthermore, other probability distribution types apart from the log-normal distribution could be considered. Lastly, no clear recommendations were made for how narrow or broad the probability distribution should be (or for how this information could potentially be derived from experimental data), which has entailed the use of rule-of-thumb estimates such as 0.1 for the standard deviation of $\log(\{\dot{p}_{Am}\})$ (Galic et al., 2017; Martin et al., 2012), as a common practice.

In this study, it was our aim to use experimental data on life-history observables of the copepod *Nitocra spinipes* reared under control conditions to make realistic estimates of the variability in DEB parameter values for this species. A thorough parameter estimation for *N. spinipes* was recently performed in DEBtool (Koch and De Schampheleere, 2019). A DEB-IBM for this copepod is currently under development in our group and is expected to strongly benefit from a realistic implementation of inter-individual variability. For the sake of model simplicity, alike the approach chosen by Martin et al. (2012), we sought to add variability to only one, i.e., the most suited parameter. The goals of this study were (a) to identify the parameter that, if sampled from a suitable probability distribution while keeping all other parameters constant, would reproduce the variation observed in the experimental data with the highest accuracy, (b) to determine the most appropriate probability distribution for this parameter to be sampled

from, and (c) to estimate the optimal scale parameter value for this probability distribution from the data.

2. Materials and methods

2.1. Model organism and DEB parameters

As a model organism, we selected the harpacticoid copepod *N. spinipes*. Preferring mainly brackish environments, this euryhaline bottom dweller can be found in different water bodies around the world (Lang, 1948). As part of the meiofauna, *N. spinipes* serves as a trophic link between the microphytobenthos and higher organisms such as fish (Coull, 1990). In the past, this species has been used frequently in ecotoxicological studies and has become the subject of multiple standardized testing guidelines (ISO, 1999, 2016; OECD, 2014). An empirical IBM for this species was firstly developed by Preuss et al. (2011) and was expanded by Koch et al. (2017). A DEB model, based on differential equations describing the life history of individuals (specifically females) more mechanistically while adhering to the law of conservation of energy, was parametrized by Koch and De Schampheleere (2019). In fact, two slightly different models were parametrized, each making different assumptions about the metabolic behavior of *N. spinipes*. The first model, denoted ‘sbp’, which stands for standard (as in the standard DEB model) metabolic behavior from birth to puberty, assumes the widely prevalent von Bertalanffy growth pattern from birth until puberty whereas the model denoted ‘abp’ assumes a metabolic acceleration (see Kooijman, 2014), which has been found in some closely related species. A defining property of both models is that unlike with the standard DEB model, the animal stops growing at puberty and subsequently all available energy that is not required for maintenance is invested in reproduction. A full definition of the abp and sbp models for copepods can be found in the comments document for the DEB book (Kooijman, 2010a). Despite their intrinsic difference, both models showed reasonable fits when parametrized individually to the same set of experimental data by Koch and De Schampheleere (2019). Therefore, the methods presented in this study were applied to both models. While in this manuscript we only report our work with the sbp model, repeating the calculations with the abp model led to very similar results, which can be found in the supplementary material. The 12 primary DEB parameters of the sbp model can be found in Table 1. It should be noted that the parameters $\{\dot{F}_m\}$, κ_X , κ_R , and k_J were not estimated from experimental data by Koch and De Schampheleere (2019) but instead were fixed at typical (“default”) values. This approach is commonly used if the available data do not contain sufficient information to make reliable estimates for certain

Table 1

Primary parameters of the sbp-type DEB model optimized to life-history data of *N. spinipes* (females) by Koch and De Schampheleere (2019).

Symbol	Unit	Interpretation	Value
$\{\dot{F}_m\}$	$L d^{-1} cm^{-2}$	Specific searching rate	6.5
κ_X	–	Assimilation efficiency	0.8
$\{\dot{p}_{Am}\}$	$J d^{-1} cm^{-2}$	Surface-area-specific maximum assimilation rate	48.17
$\dot{v}$	$cm d^{-1}$	Energy conductance	0.0239
κ	–	Allocation fraction to soma	0.610
κ_R	–	Reproduction efficiency	0.95
$[\dot{p}_M]$	$J d^{-1} cm^{-3}$	Volume-specific somatic maintenance rate	817.37
$\{\dot{p}_T\}$	$J d^{-1} cm^{-2}$	Surface-area-specific somatic maintenance rate (typically zero in ectotherms)	0
k_J	d^{-1}	Maturity maintenance rate coefficient	0.002
$[E_G]$	$J cm^{-3}$	Volume-specific costs for structure	4454.72
E_H^b	J	Maturity level at birth	6.12×10^{-4}
E_H^p	J	Maturity level at puberty	8.18×10^{-2}

parameters (Marques et al., 2018). Moreover, the surface-area-specific somatic maintenance rate $\{\dot{p}_T\}$ was fixed at zero, which is typically done for ectotherms (Kooijman, 2010b).

2.2. Experimental data and model simulations

For the assessment of variability of life-history traits in *N. spinipes*, we made use of data from multiple life-history experiments of very similar design that were carried out in the past (Dahl and Breitholtz, 2008; Koch et al., 2017; Lundström et al., 2010; OECD, 2007; Preuss et al., 2011). Data on two endpoints, i.e., (a) the time from hatching until reaching the reproductive state, hereafter referred to as “development time” (sample size $n = 610$) and (b) the number of offspring per brood of a reproducing female, hereafter referred to as “brood size” ($n = 606$), were deemed fit for purpose since their sample sizes were considered large enough to provide insight into their respective distributions’ shape. Moreover, both endpoints are closely linked to the population growth rate and are thus of high ecological relevance (Hansen et al., 1999). All data on animals that had been exposed to chemical stressors were filtered out for this study. While not all studies used the same food source, food was always offered in abundance. Due to the relatively small size of the experimental setups (in microwell plates), high consistency in exposure temperatures among individuals from the same experiment could be ensured. This gives us reason to believe that inter-individual variation observed within each experiment was mainly caused by physiological differences among the animals instead of differences in their environment. To account for minor temperature differences between the experiments, all data were normalized. This was done by dividing all data points by the mean of their respective data set and subsequently multiplying them by the average development time or brood size at optimal conditions of temperature (20°C) and food (functional response $f = 1$), predicted by the model. Density histograms of the experimental data are presented in Fig. 1.

In the present study, all model simulations of the endpoints development time and brood size at different parameter values were done using the functions provided in the free software package DEBtool (<http://www.bio.vu.nl/thb/deb/deblab/>). Since the DEB model keeps track of the maturity energy E_H (i.e., the cumulative energy, which has been invested by the organism to build up maturity) over time, the development time from birth ($E_H = E_H^b$) until puberty ($E_H = E_H^p$) is an endpoint that could be directly predicted by the DEB model. The reproduction rate $\dot{R}$ of a reproducing female was calculated via $\dot{R} = \frac{\kappa_R \times \dot{p}_R}{E_0}$ with the reproduction efficiency κ_R (see Table 1), the energy flux to reproduction $\dot{p}_R$ (which in the sbp-type model at puberty equates to the assimilation flux $\dot{p}_A$ minus the energy fluxes to somatic maintenance $\dot{p}_S$ and maturity maintenance $\dot{p}_T$), and the energy cost per egg E_0 . Consequently, reproduction output was predicted as the number of offspring per day instead of offspring per brood. Therefore, additional information about the timing per brood would be needed to properly link the model output to the experimental data. Since this information was missing for most of the brood size data, we assumed a constant release time of 3.17 days per brood, i.e., the average of all the available data. Unfortunately, this assumption unavoidably led to more uncertainty when linking the model outputs to the data. For this reason, we decided not to include the brood size data when fitting parametric distributions to the data in subsection 2.4 but instead use it for preliminary model validation.

2.3. Parameter screening

Our objective was to identify individual model parameters that, if drawn from an appropriate probability distribution while keeping all other parameters at their nominal values (as in Table 1), would produce variation in the model output that is a good approximation of the

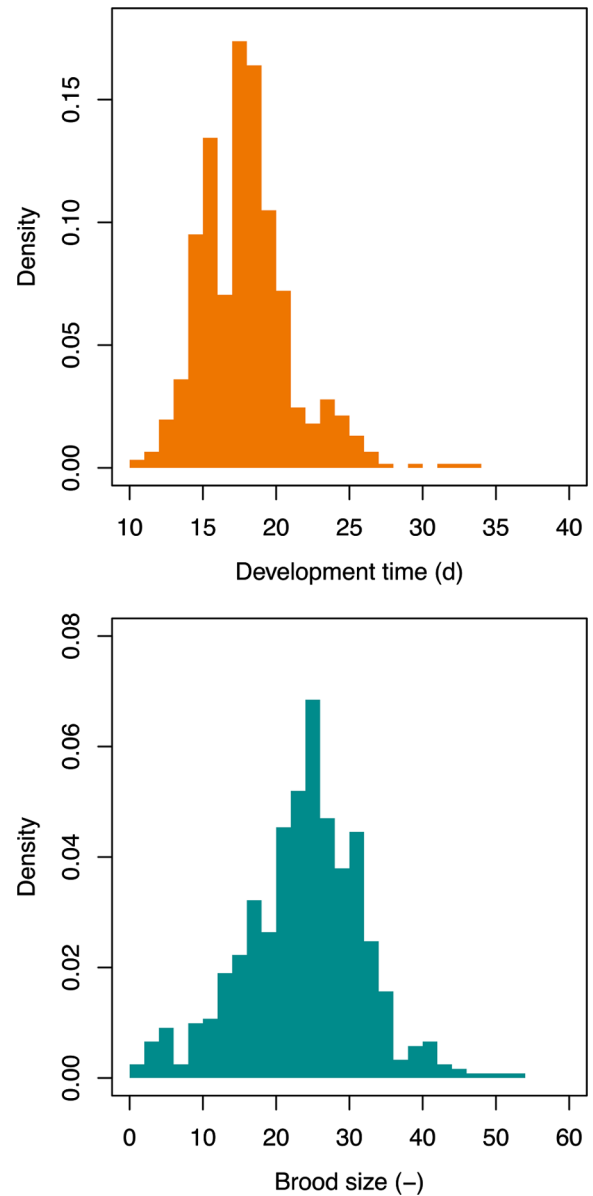


Fig. 1. Density histograms of experimental data on the development time until puberty (top; $n = 610$) and brood size per reproducing female (bottom; $n = 606$) of *N. spinipes*. All data were normalized to optimal conditions of temperature and food as described in Section 2.2.

variation that was observed in the experimental data. As the first step in the search for such a parameter, a one-at-a-time sensitivity analysis (OAT) was performed. In accordance with the experimental conditions, the model simulations were executed under optimal conditions of temperature (20°C) and food (functional response $f = 1$). Each parameter from Table 1 (for the exceptions, see below) was moved stepwise from extremely low to extremely high values while the produced model outputs for development time and brood size were recorded in each step. Extreme parameter values here are defined as (a) parameter boundaries such as 0 and 1 for ratio parameters, (b) values around which the model outputs converged toward stable values, or (c) parameter values below or above which simulations failed due to unrealistic parameter combinations. For each parameter that was moved between its extreme values, the range of model output data on development time and brood size was then compared to the experimental data. Only if the range of model output data encompassed the 5th and 95th percentile of the experimental data, a parameter was considered a plausible

candidate to explain the variation observed in the experimental data. In the case of the parameters E_H^b and E_H^p , which are the maturity thresholds for birth and puberty, respectively, we supposed proportionality. These parameters indicate how much cumulative energy needs to be invested by the organism in maturity before birth (i.e., the moment the organism starts feeding) and, later on, puberty is reached. Inter-individual variations in these two parameters may be explained by differences in how efficiently organisms can use energy for maturation. We assumed that individuals that require above average energy investment in maturity to reach birth, also need more energy to reach puberty. Therefore, these two parameters were not moved individually but instead were multiplied by a newly introduced factor x_{EH} , which in turn was moved from low to high values around a default of 1. In further analyses, x_{EH} itself (instead of E_H^b and E_H^p individually) was treated as a (potentially variable) DEB parameter. Recent studies have suggested that multiple DEB parameters may covary among individuals of the same species following the same general rules that govern covariation in (mean) parameter values among different species (Stavrakidis-Zachou et al., 2019). These inter-specific rules, also referred to as body size scaling relationships, have been discussed in great detail in the past (Kooijman, 2010b). Carrying over these rules from the inter-species to the intra-species level implies that when $\{\dot{p}_{Am}\}$ is modified (relative to its default value) by a factor ζ , both E_H^b and E_H^p must be multiplied by ζ^3 . Notably, this approach also assumes proportionality between E_H^b and E_H^p . To further explore this idea, we also included ζ (which modifies $\{\dot{p}_{Am}\}$, E_H^b , and E_H^p simultaneously) in our parameter screening. Furthermore, some parameters were excluded from the parameter screening: Firstly, the specific searching rate $\{\dot{F}_m\}$, which is used to calculate the animal's feeding rate as a function of food density, was excluded since high food concentrations were present in all experiments and a functional response of $f = 1$ was assumed for all simulations. Strictly speaking, despite being defined as a primary DEB parameter, the maximum assimilation rate $\{\dot{p}_{Am}\}$ is the product of the maximum ingestion rate $\{\dot{p}_{Xm}\}$ (not a primary parameter) and the assimilation efficiency κ_X . Hence, potential variability in κ_X is also contained in $\{\dot{p}_{Am}\}$. Since the data used in this study did not allow to differentiate between variability in $\{\dot{p}_{Xm}\}$ versus κ_X (as the source for variability in $\{\dot{p}_{Am}\}$), κ_X was excluded from the screening. Likewise, the reproduction efficiency κ_R was excluded since it does not affect any processes other than reproduction. Lastly, the surface-area-specific somatic maintenance rate $\{\dot{p}_T\}$, which is typically zero for ectotherms (Kooijman, 2010b), was also excluded from the parameter screening.

2.4. Fitting parametric distributions

As a next step, we aimed to find an appropriate parametric probability distribution for each of the parameter candidates, which passed the screening, to be sampled from. We used the curves generated in the OAT (Fig. 2) to deduce the exact values of each parameter candidate that would be needed to replicate the experimental data on development time, one-by-one, in model simulations. The thus obtained parameter distributions were then analyzed by means of skewness-kurtosis plots (Cullen and Frey, 1999) ('descdist' function in R package 'fitdistrplus'). Such plots make use of an empirical distribution's skewness and kurtosis as two shape-defining properties to investigate its similarity to commonly used parametric distributions. The idea behind this is that the closer the distance between the empirical distribution and a parametric one, in a plot of kurtosis versus squared skewness, the better the fit that can be expected. The most plausible parametric distributions were then fitted to the data by means of the 'fitdist' function in the same R package. Lastly, Akaike information criteria (AIC) were used to identify the most suited parametric distribution for each parameter.

2.5. Model implementation and validation

In this final step, variability was added to the DEB model for *N.*

spinipes by, for each new simulation, drawing one particular model parameter from its optimal probability distribution and keeping all other parameters fixed. This procedure was implemented separately for all parameters that had passed the parameter screening in Section 2.3. To test the different implementations, for each one, 10^5 life-history simulations were executed while development time and brood size were recorded as endpoints. The thus obtained distributions of both endpoints were then compared to the experimental data distributions in consideration of their shape and scale. As the development time data had been used to fit the parameter distributions, a comparison of simulations versus the experimental data should only serve as a sanity check for the fitting method. The brood sizes, in contrast, had not been included in the fitting process and were used as an "independent" data set to validate the model.

3. Results and discussion

3.1. Parameter screening

The results of the parameter screening are presented in Fig. 2. Besides the simulation data from each parameter in the OAT, the figure shows the range of experimental data (full range as well as 5th and 95th percentiles). Overlaps are shaded in light blue whereas discrepancies are shaded in pink. It can be seen that $\{\dot{p}_{Am}\}$ and x_{EH} were the only parameters that if varied while keeping all other parameters constant could reproduce the 5th–95th percentile range of measured data on both development time and brood size. Although changes in $[E_G]$ could explain the range of measured development times, they could not explain the range of brood sizes. By contrast, varying $\{\dot{p}_M\}$ led to the observed brood sizes but the lower 5% of development time data could not be reached by changes in this parameter alone. All other parameters failed to reproduce short development times and high brood sizes even if changed (individually) to extremely low or high parameter values. This also includes the factor ζ , which modifies $\{\dot{p}_{Am}\}$, E_H^b , and E_H^p simultaneously, following the principle of body size scaling relationships. Therefore, we conclude that, despite its applicability on the inter-species level, this principle is not suitable to explain all variation observed on the intra-species level. Ultimately, only $\{\dot{p}_{Am}\}$ and x_{EH} were left as parameters to move forward with.

3.2. Fitting parametric distributions

Hypothetical distributions of the parameters $\{\dot{p}_{Am}\}$ and x_{EH} , that, with all other parameters fixed, would be needed to accurately replicate the experimental development time data, are presented in Fig. 3. Both distributions are presented as points in a plot of kurtosis versus squared skewness to illustrate their proximity to commonly used parametric probability distributions, which, depending on the shapes they can take, are displayed as points, lines, or areas. The distribution of x_{EH} is distinctly skewed and may presumably be approximated by distributions that allow for asymmetry such as the log-normal, the gamma, the exponential, and the beta distribution. The latter, however, is confined to the range (0, 1) and, therefore, was excluded. The distribution of $\{\dot{p}_{Am}\}$ is much less skewed and may also be approximated by symmetric distributions such as the normal and the logistic distribution. An overview of all conducted distributions' fits with goodness-of-fit criteria can be found in the supplementary material.

For both distributions, a log-normal distribution was found to give the best fit, closely followed by the gamma distribution. We consider both these theoretical distributions appropriate approximations of the parameter distributions of $\{\dot{p}_{Am}\}$ and x_{EH} . For this study, we decided to proceed with the log-normal distribution due to its slightly better fit. Whereas both parameters' distributions could be approximated by the same theoretical distributions, the distributions' scale parameters were found to be distinctly different. With a standard deviation of 0.40, the scale of x_{EH} 's log-normal distribution is wider than that of $\{\dot{p}_{Am}\}$

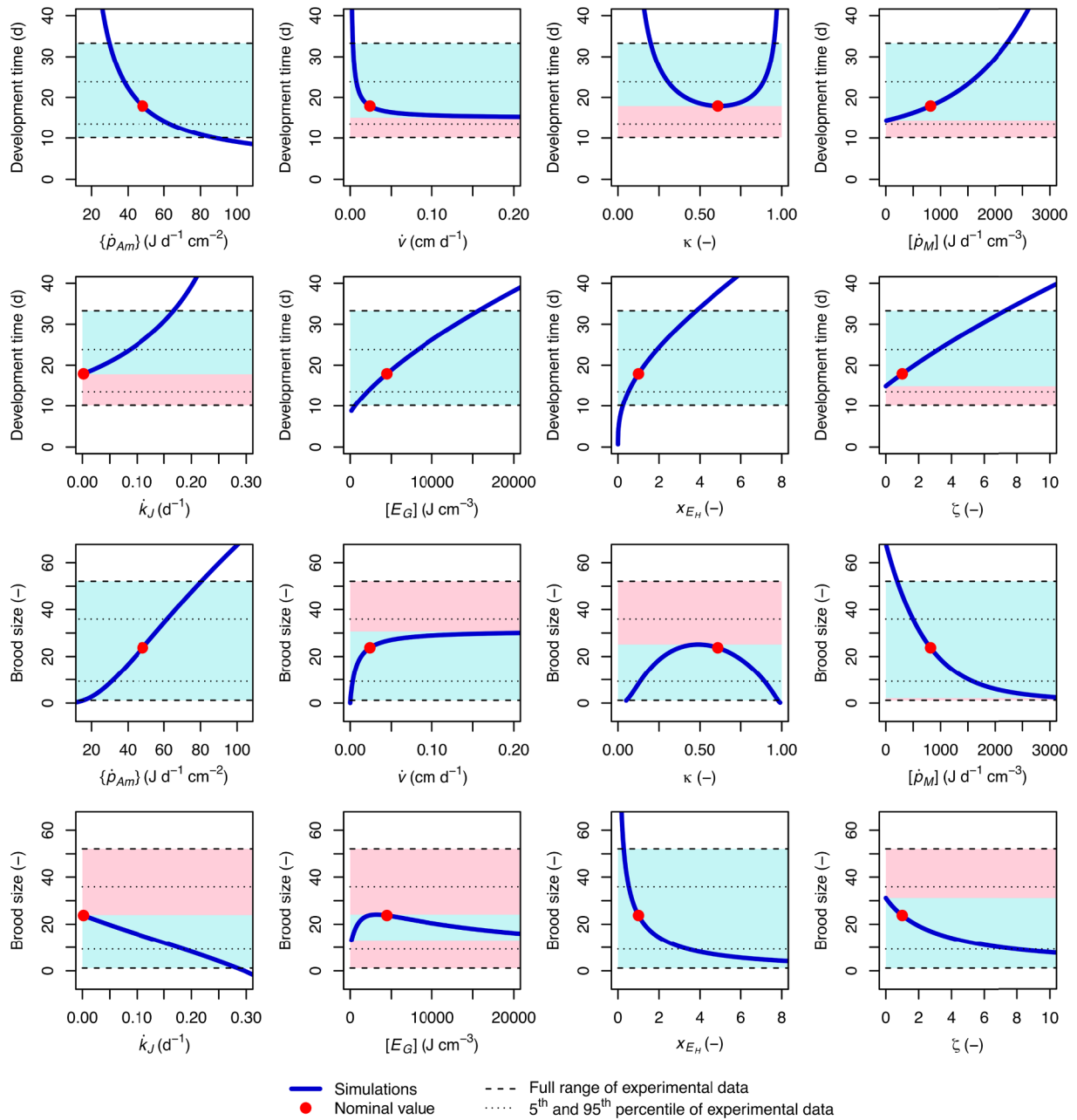


Fig. 2. One-at-a-time sensitivity analysis of the DEB parameters for *N. spinipes*. Simulated data on development time and brood size (blue lines) per parameter are contrasted with the range of experimental data (areas between dashed lines). Overlaps are shaded in light blue whereas discrepancies are shaded in pink. Red dots indicate the point at which all parameters are at their nominal value as displayed in Table 1. The factor x_{EH} modifies both parameters E_H^b and E_H^p proportionally. The factor ζ , by analogy with body size scaling relationships (Kooijman, 2010b), modifies $\{p_{Am}\}$, E_H^b , and E_H^p simultaneously as described in Section 2.3. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

($SD = 0.15$), meaning that larger relative changes in x_{EH} are needed to produce the same changes in the model output. At this point, we can not make any judgment about what variability in each respective parameter is biologically reasonable or not. Additional case studies on variability in DEB parameters, which consider multiple species and endpoints, should be performed in the future to develop a broader understanding of this matter.

3.3. Model implementation and validation

Results of 10^5 repeated model simulations with the best fitting parameter distributions of $\{p_{Am}\}$ and x_{EH} , respectively, are presented in Fig. 4. Histograms of the experimental data (same as in Fig. 1) are

shown for comparison. Notably, regardless of whether $\{p_{Am}\}$ or x_{EH} was subject to variation, the simulation results for the considered observables were virtually identical. A closer inspection revealed close-to-perfect linearity ($R^2 = 0.998$) between the natural logarithm of the values of $\{p_{Am}\}$ and x_{EH} that lead to the same output in development time within the range of realistic, i.e., experimentally measured values. Hence, drawing both parameters from log-normal distributions, fitted to the same data, could be expected to deliver equivalent results in the model outputs for this endpoint. The fact that the same phenomenon also appears to be true for the brood sizes points towards a close mechanistic link between the two endpoints in the model structure. For further elaborations see Section 3.4.

The distributions of simulated development times correspond well

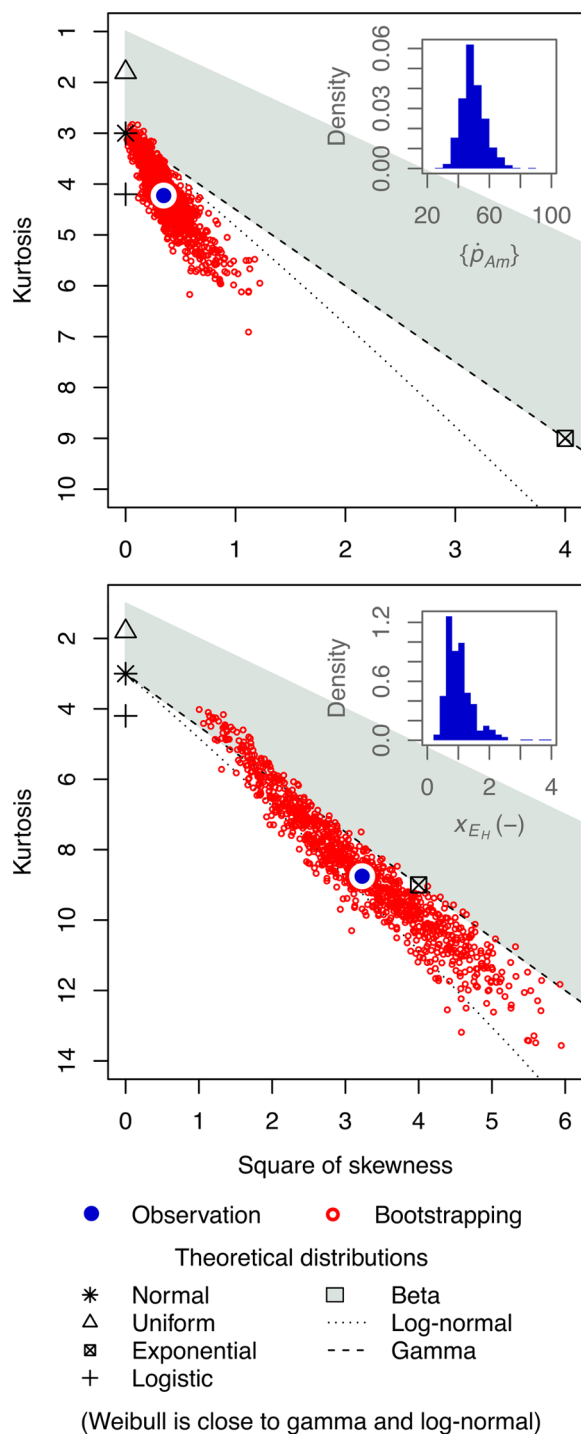


Fig. 3. Skewness-kurtosis plots of the distributions of parameters $\{p_{Am}\}$ (top) and x_{EH} (bottom) that would be needed to perfectly replicate the experimental data on development time (if all but the parameter in question remain at their nominal value). These plots indicate each distribution's skewness and kurtosis as two defining measures of its overall shape to illustrate their proximity to common parametric distributions. The blue dots indicate each respective distribution's position in the skewness-kurtosis space. Red dots represent bootstrap samples of these same distributions. Theoretical probability distributions are marked as symbols, lines, or shaded areas, depending on the shapes they can take. Additionally, density histograms of both parameter distributions are presented in the upper right corner of each panel. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

with the distribution of the experimental data both in terms of shape and scale. While development times of 15, 17, and 18 days were slightly less frequent (in relation to the full sample) in model simulations versus the experimental data, development times of 16, 21, and 22 days occurred slightly more often. This can likely be explained by the limited sample size of the experimental data as we would expect a more smooth density distribution at a higher n . The overall convincing fit of simulated versus experimental data for this endpoint could be expected since the parameter distributions had been fitted (although indirectly) to the data. Next, when looking at brood size, the simulated data again correspond fairly well with the experimental data in terms of their spread. The model, however, failed to reproduce the left-sided skewness of the experimental data. By sampling either of the parameters $\{p_{Am}\}$ or x_{EH} from a log-normal distribution, the produced model output for brood size suggests to follow a distribution similar to the log-normal, which is likewise right-skewed. It can be noted that the experimental data do not follow a perfectly smooth distribution but instead have a little hump at particularly small brood sizes (between 1 and 6). From the curvilinear relationships of individually varied parameters versus the model outputs in the OAT (Fig. 2), it could be expected that this unevenness in the brood size histogram could not be reproduced by varying only one single physiological parameter with a two-parameter probability distribution. The data suggest that more than one process may have an effect on the brood size and this cannot be fully explained by variation in one physiological parameter alone. However, it must also be noted that the experimental data, to an unknown degree, suffer from multiple sources of uncertainty such as non-ideal experimental setups, handling of the test animals, data collection, and data normalization. Furthermore, as stated above, the model presumed a constant brood release rate, which was independent of any parameter variations. It can be speculated, that in reality the release rate is positively or negatively correlated with the brood size. Good quality data on this are currently lacking. Considering the uncertainties in both the experimental data and the model output, we find the overall match of the two satisfactory. The realistic coverage of variability in both, the development time until the onset of reproduction and the reproductive output per female, makes the DEB model very useful as a component in an individual-based population model since the two are fundamental drivers of population growth (Hansen et al., 1999). Data on other life-history observables of *N. spinipes* may be collected in the future to further validate the model and, more specifically, its implementation of inter-individual variability in different life-history traits. Having access to more data, however, may also lead to a case in which not all variation observed in different endpoints can be explained well by variability in one parameter alone. Since DEB parameter values emerge from complex physiological systems, which for a large part may be variable among individuals, it seems natural that in an ideal model more than one of these parameters should be subject to some degree of variation. This hypothesis is supported by the fact that various DEB parameters also differ widely among different species in the animal kingdom, whereas others appear to be more universally consistent (Lika et al., 2011). From an evolutionary perspective, inter-individual variability of multiple DEB parameters within a species is a fundamental requirement for new species (with different DEB parameters) to emerge. That being said, depending on the scientific question that a model aims to answer, reducing the variability to just one parameter may be an appropriate simplification of reality.

3.4. Parameter considerations

Although one goal of this study was to identify a single, i.e., the most suitable parameter that should be varied to produce realistic variation in the model outputs, ultimately, the two parameters $\{p_{Am}\}$ and x_{EH} were found to be equally suitable. The explanation for this can be found in the model structure: An increase in the maximum assimilation rate $\{p_{Am}\}$, which goes along with a higher energy uptake, leads

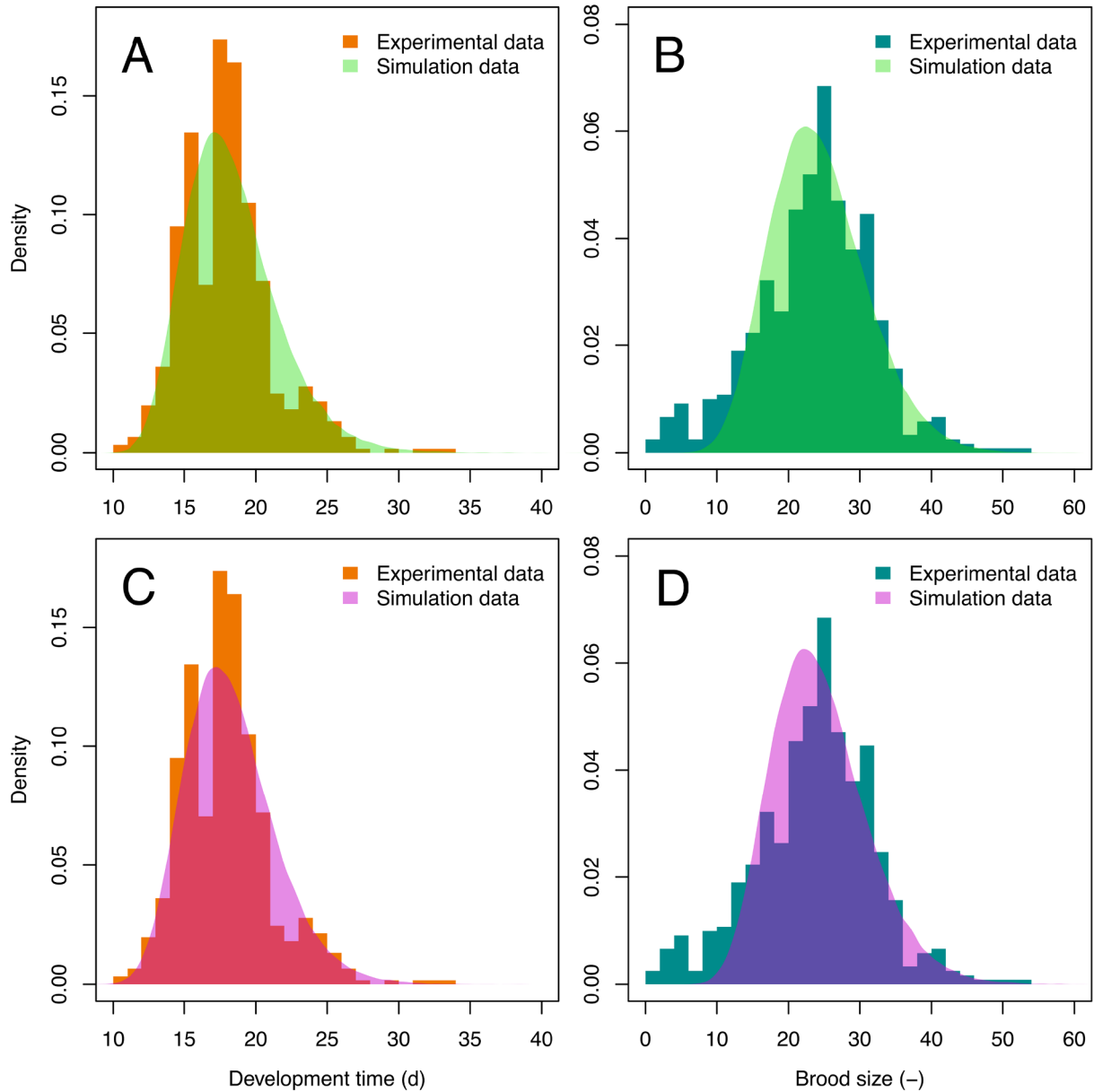


Fig. 4. Experimental data versus simulated data from the sbp-type DEB model for *N. spinipes* with either the parameter $\{p_{Am}\}$ (A and B) or x_{EH} (C and D) being randomly sampled from a log-normal distribution while all other parameters were fixed. The parameter distributions were fitted to the development time data, only. The brood size data are used to test the model's validity.

to higher energy investment in maturity and consequently an earlier achievement of the maturity threshold for puberty. After puberty, more energy is available for reproduction, which leads to the production of more offspring. The same result can be achieved by lowering x_{EH} , which translates to lowering both the maturity thresholds for birth (E_H^b) and puberty (E_H^p) by the same proportion. In this scenario, less energy is needed to reach the maturity thresholds, which again leads to faster maturation. A lower E_H^b leads to lower energetic costs per produced egg and, again, leads to the production of more offspring. The opposite is the case if $\{p_{Am}\}$ is decreased or x_{EH} is increased. Interestingly, the just described results of increasing $\{p_{Am}\}$ or decreasing x_{EH} (or vice versa) are not only equivalent in terms of quality but also (approximately) in terms of quantity: a positive change in $\{p_{Am}\}$ and a negative change in x_{EH} that would each lead to the same reduction in development time, would also each lead to the almost exact same increase in brood size. This relationship appears to be a special property of the typified copepod models abp and sbp in which growth ends at puberty, and does

not hold true, e.g., for the standard DEB model. Although at this point it may seem irrelevant for the model outputs whether $\{p_{Am}\}$ or x_{EH} is subject to variability, the situation changes if we take other endpoints into consideration. If, for example, we consider the body size at puberty as an additional model output, we find that an increase in $\{p_{Am}\}$ leads to a larger body size whereas a decrease in x_{EH} leads to a smaller one, despite the similar development time and brood size per female. This may have important ecological implications when considering factors such as food competition (as larger animals consume more food) and size-selective predation, which should be further explored in future simulation studies. Overall, variation in either $\{p_{Am}\}$ or x_{EH} appears ecologically meaningful since they both promote the desynchronization of a population in terms of its life stages, which in turn makes it less vulnerable to environmental stress and catastrophic mortality events that affect only specific fractions of the population (Ohlberger and Langangen, 2015). Additional experimental work, investigating correlations between either the development time or brood size on the one

hand, and the body size at puberty, on the other hand, could inform the prioritization of one parameter over the other when it comes to the implementation of variability in the model. For now, we prefer the use of $\{\dot{p}_{Am}\}$ as an individually variable parameter over x_{EH} , since using x_{EH} as a surrogate parameter for E_H^b and E_H^p is slightly more cumbersome. Furthermore, it is based on the untested model assumption that E_H^b and E_H^p covary proportionally among individuals, which should be further explored in the future.

3.5. Extrapolation to other species

A striking convenience of the DEB theory is that it is highly generic in the sense that virtually any species' life history can be reproduced by the same model or slightly modified derivatives thereof (Kooijman, 2010a). The differences between two species are primarily explained by differences in their parameter values. Dealing with intra-specific variability of DEB parameters raises the question if it is transferable from one species to another. It can be assumed that individuals of closely related species vary in the same parameters. The extent to which parameters vary may, however, be highly heterogeneous and likely depends on a species' ecology. Furthermore, the role and interpretation of a DEB parameter can be slightly different when comparing, e.g., the standard DEB model to a model derivative such as abp or sbp. Nonetheless, it can be noted that the optimal approach found in this study, i.e., varying $\{\dot{p}_{Am}\}$ with a log-normal distribution, coincides with the approach chosen in earlier studies (Galic et al., 2017; Martin et al., 2012). We, therefore, assume that the existing approach, although to our knowledge not based on real observations but rather implemented as a pragmatic solution, may in fact be quite accurate. At 0.15, the ideal standard deviation of $\log(\{\dot{p}_{Am}\})$ calibrated in this study was slightly higher than the commonly applied default of 0.1 but, again, we expect this value to differ on a species-to-species base. This is also why we advise against one-to-one extrapolations of our results to other species unless this can be supported with experimental data. Nevertheless, the methods presented in this study may certainly be applied to other species for which similar data exist. This study should, first and foremost, be seen as a proof of principle for how the very complex question of intra-specific variability in a DEB model can be addressed in a simple and straightforward manner.

4. Conclusions

In this study we presented a novel approach for how the inter-individual variability of DEB model parameters can be estimated from experimental data, using the copepod *N. spinipes* as a model species. The initial choice to add variability to only one model parameter while keeping all others at their default greatly facilitated dealing with this otherwise very complex task. Yet, the "one-parameter approach" delivered satisfactory results. By 'recycling' experimental data from previous studies, we were able to find the most plausible parameters that may be subject to variability, to identify suitable parametric probability distributions, and to fit those distributions to the data. In the end, two parameters were found to be equally suitable to reproduce the variability observed in the experimental data. Additional data in the future may tell which of the two parameters is the most biologically plausible. The variability of two life-history traits that are closely related to population growth was predicted well by the model, supporting its potential use in IBMs. Although we advise against extrapolating the results of this study to other DEB organisms one-to-one, we hope that the methods established in this study inspire other researchers to carry out similar studies with different species.

CRediT authorship contribution statement

Josef Koch: Conceptualization, Methodology, Software, Formal

analysis, Investigation, Writing - original draft, Visualization. **Karel A.C. De Schampelaere:** Conceptualization, Investigation, Resources, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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